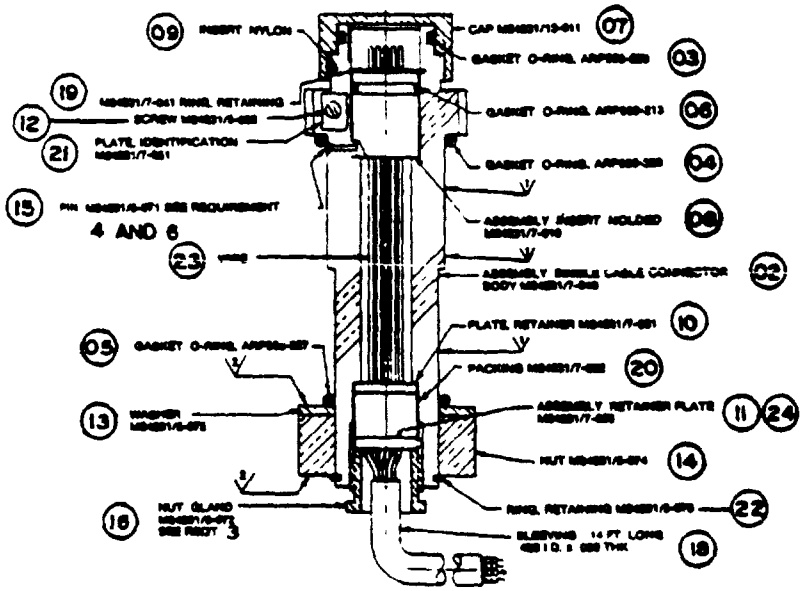
MIL-C-24231/7E(SH)
15 May 1987
SUPERSEDING
MIL-C-24231/7D(SHIPS)
30 August 1976
(See note 4)

MILITARY SPECIFICATION SHEET

CONNECTOR, SINGLE CABLE, TYPE III, SEVEN-CONDUCTOR

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the single cable connector described herein shall consist of this specification and the latest issue of MIL-C-24231.



1/ Hardness reading to be taken at these points.

2/ Hardness reading to be taken at this face at 120-degree intervals.

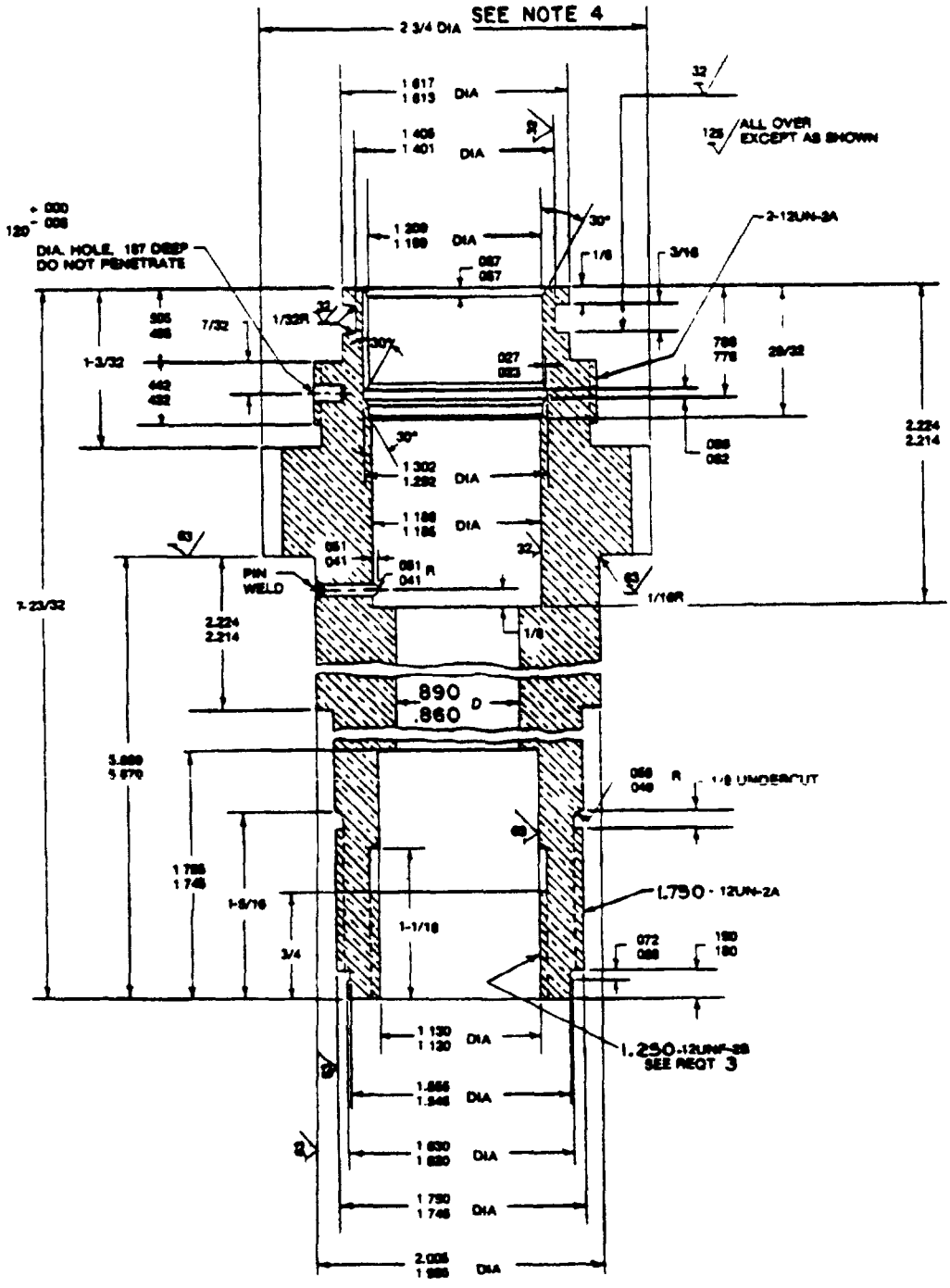
SH 9674

FIGURE 1. Connector, seven-conductor - M24231/7-001, symbol no. 517.1. (01)

AMSC N/A FSC 5935

DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

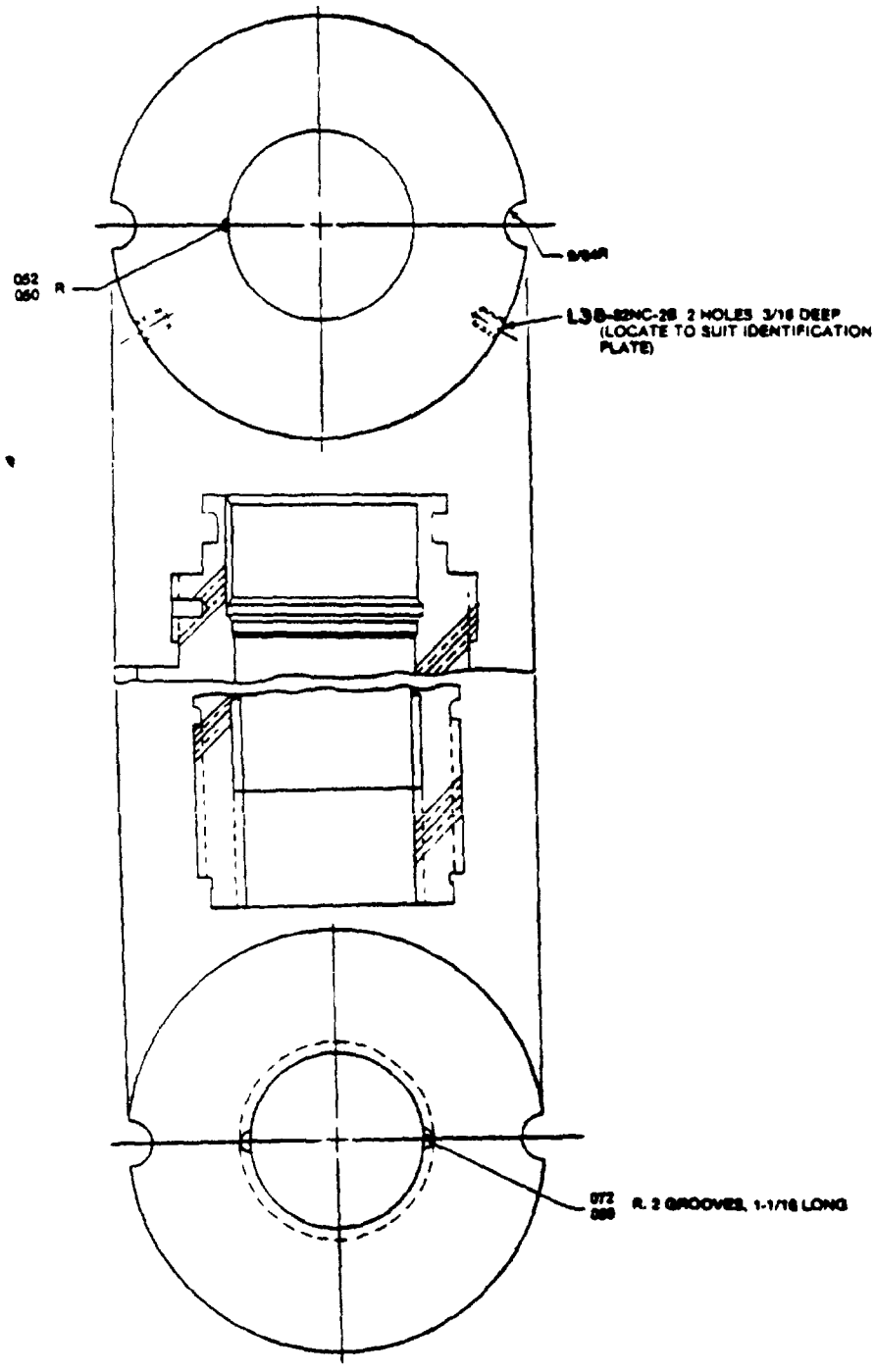
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SH 9674-1

FIGURE 2. Assembly, single cable connector body - M24231/7-040. (02)

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SH 9674-2

FIGURE 2. Assembly, single cable connector body - M24231/7-040. - Continued (02)

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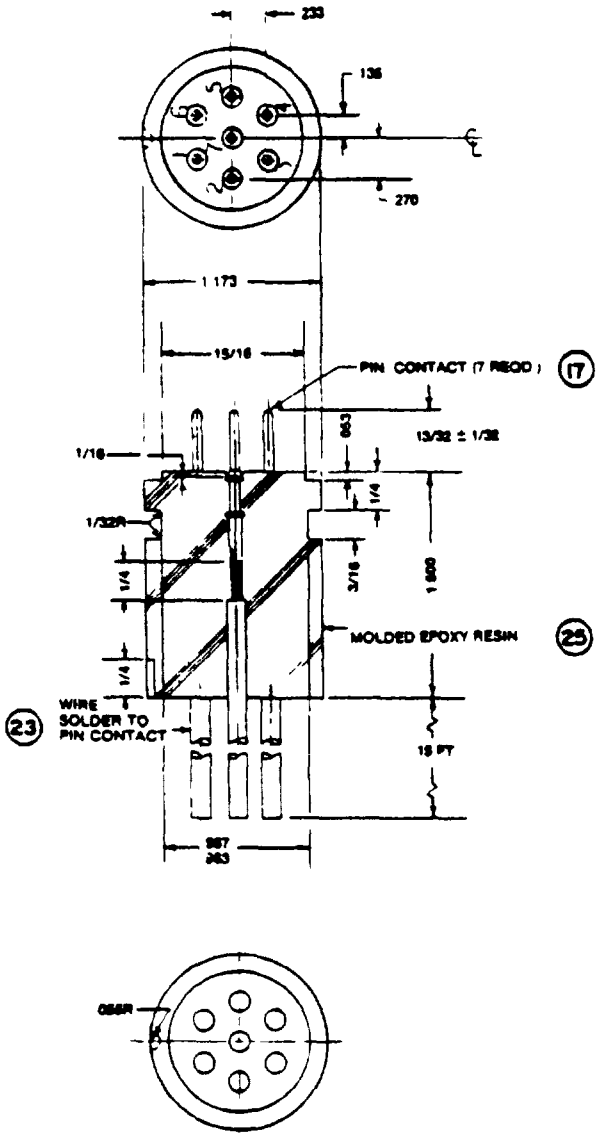


FIGURE 3. Assembly, insert, molded - M24231/7-010. 08

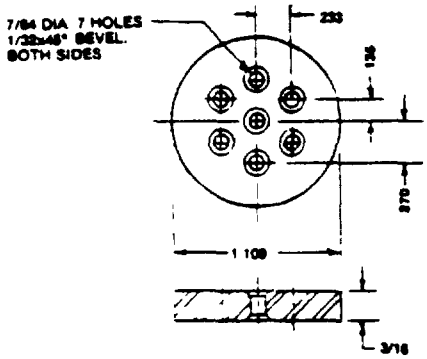


FIGURE 4. Plate, retainer - M24231/7-031. 10

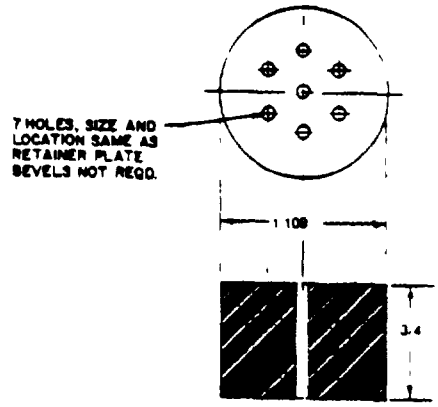


FIGURE 5. Packing - M24231/7-032. 20

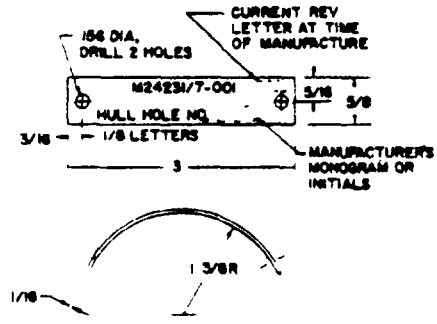


FIGURE 6. Plate, identification - M24231/7-051. 21

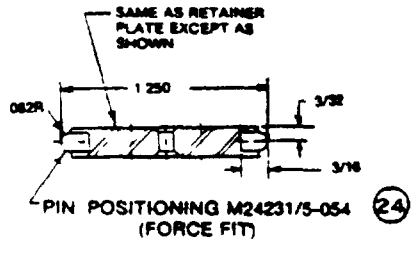


FIGURE 7. Assembly, retainer plate - M24231/7-030. 11

SH 9674-3

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REQUIREMENTS:

1. Dimensions and configuration: See figures.
2. Full or partial threads shall not extend more than 0.083 inch beyond the 3/4-inch thread length.
3. The gland nut shall be tightened until the packing tightens against the wires. This can be determined by moving the wires gently back and forth until the wire is seized by the packing. The gland nut shall then be turned approximately one-fourth of a turn more to hold the wires in place. Do not overtighten the gland nut or attempt to tighten down metal to metal, as this can damage the wires.
4. Refer to requirement 4 of MIL-C-24231/5 for alternate method of manufacturing connector body and installing guide pin M24231/5-071.
5. The retainer plates and retainer plate assemblies shall be manufactured from nickel-copper alloy in accordance with class A, form 1, hot finished condition of QQ-N-281 with a minimum yield strength of 40,000 lbf/in^2 at 0.2 percent offset, or nickel-copper alloy in accordance with class A, form 1, cold drawn condition of QQ-N-281 with a minimum yield strength of 50,000 lbf/in^2 at 0.2 percent offset.
6. See requirement 7 of MIL-C-24231/5.

TABLE I. Parts list.

Item 1/ no.	Description 2/	3/ Quantity for M24231/7- 001	Material/specification	Remarks
01	Connector, assembly, single cable, symbol no. 517.1	1	-----	See figure 1
02	Connector, body assembly, single cable	1	Ni-Cu-Al, QQ-N-286	See figure 2
03	Gasket, O-ring	1	MS28775-220	Lubricate with silicone compound, MIL-S-8660
04	Gasket, O-ring	1	MS28775-329	See item 03
05	Gasket, O-ring	1	MS28775-327	See item 03
06	Gasket, O-ring	1	MS28775-213	See item 03
07	Cap, pressure-proof	1	M24231/13-011	See figure 1
08	Insert assembly, molded	1	See items 17, 23, and 25	See figure 3

See footnotes at end of table.

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TABLE I. Parts list. - Continued

Item 1/ no.	Description 2/	3/ Quantity for M24231/7- 001	Material/specification	Remarks
09	Insert	1	M24231/5-077	See figure 1
10	Plate, retainer	1	Ni-Cu, class A, QQ-N-281	See figure 4 and requirement 5
11	Assembly, retainer plate	1	Ni-Cu, class A, QQ-N-281	See figure 7 and requirement 5
12	Screw	2	M24231/5-083	See figure 1
13	Washer	1	M24231/5-075	See figure 1
14	Nut	1	M24231/5-074	See figure 1
15	Pin, guide	1	M24231/5-071	See figure 1
16	Nut, gland	1	M24231/5-072	See figure 1
17	Pin, contact	7	MIL-C-24231	See figure 3
18	Sleeving, 0.430 id, 0.030 thick	14 ft	Type F, form U, category 1, color clear, MIL-I-631	See figure 1
19	Ring, retaining	1	SPIROLOX PRB-131-M, or equal	See figure 1
20	Packing, class A, durometer 55 ± 5	1	Rubber, ZZ-R-768	See figure 5
21	Plate, identification	1	CRES or Al, type I, MIL-P-15024	See figure 6
22	Ring, retaining	1	M24231/5-076	See figure 1
23	Wire, 15 feet	7	Type B-16, MIL-W-16878, nylon jacket, black	See figures 1 and 3
24	Pin, positioning	1	M24231/5-054	See figure 7
25	Epoxy resin, molded	As required	Cast epoxy resin compound 4/	See figure 3

See footnotes at top of next page.

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1/ Item numbers are the same as encircled numbers shown on figures.
2/ See applicable figure or table listed in remarks column for part numbers and dimensions.
3/ Quantity is for one complete assembly of each size.
4/ Arcoset No. 200, M.M. Reif and Co., Rockledge, PA, or Phillicast No. 2, Philadelphia Resin Corp., Montgomeryville, PA, or equal.

NOTES:

1. All dimensions are in inches.
2. Unless otherwise indicated, tolerances shall be as follows:

   (a) Decimals - plus or minus 0.005 inch.
   (b) Fractions - plus or minus 1/64 inch.
   (c) Angular dimensions - plus or minus 1/2 degree.

3. The flange outside diameter is intended to coincide with commonly available 2-3/4 inch bar stock. The dimension may vary after machine clean-up.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5935-N279-07)

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